

Introduction to AI & LLM Applications

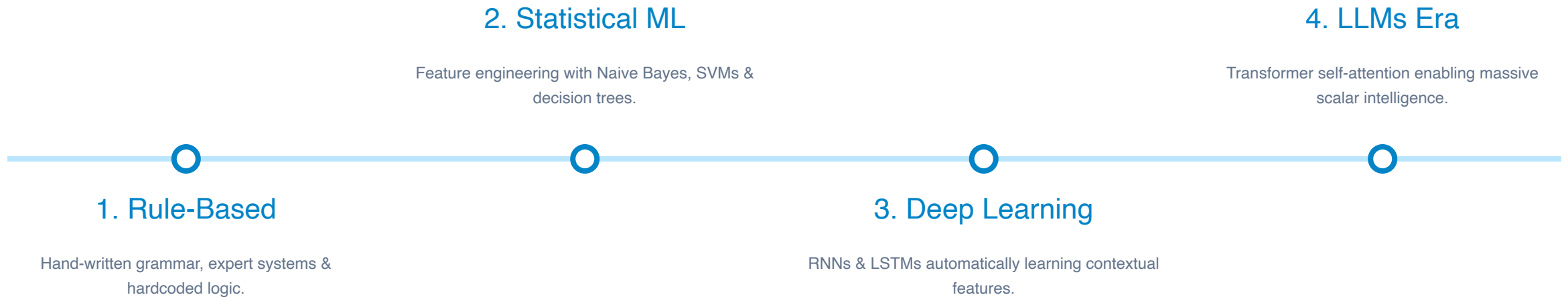
Course: GenAI, LLM and It's Foundations— A Top-Down Approach

By Dr. Kaveri Kale

14-Week Top-Down Curriculum



The Evolution of AI Systems



What Are GenAI & LLMs?

Generative AI

AI systems designed to **create original content** (text, images, code, audio) rather than strictly predicting labels or categories.

Traditional ML: Predicts spam vs. non-spam or credit risk scores.

Generative AI: Writes complete emails, generates working code, or drafts legal summaries.

Large Language Models

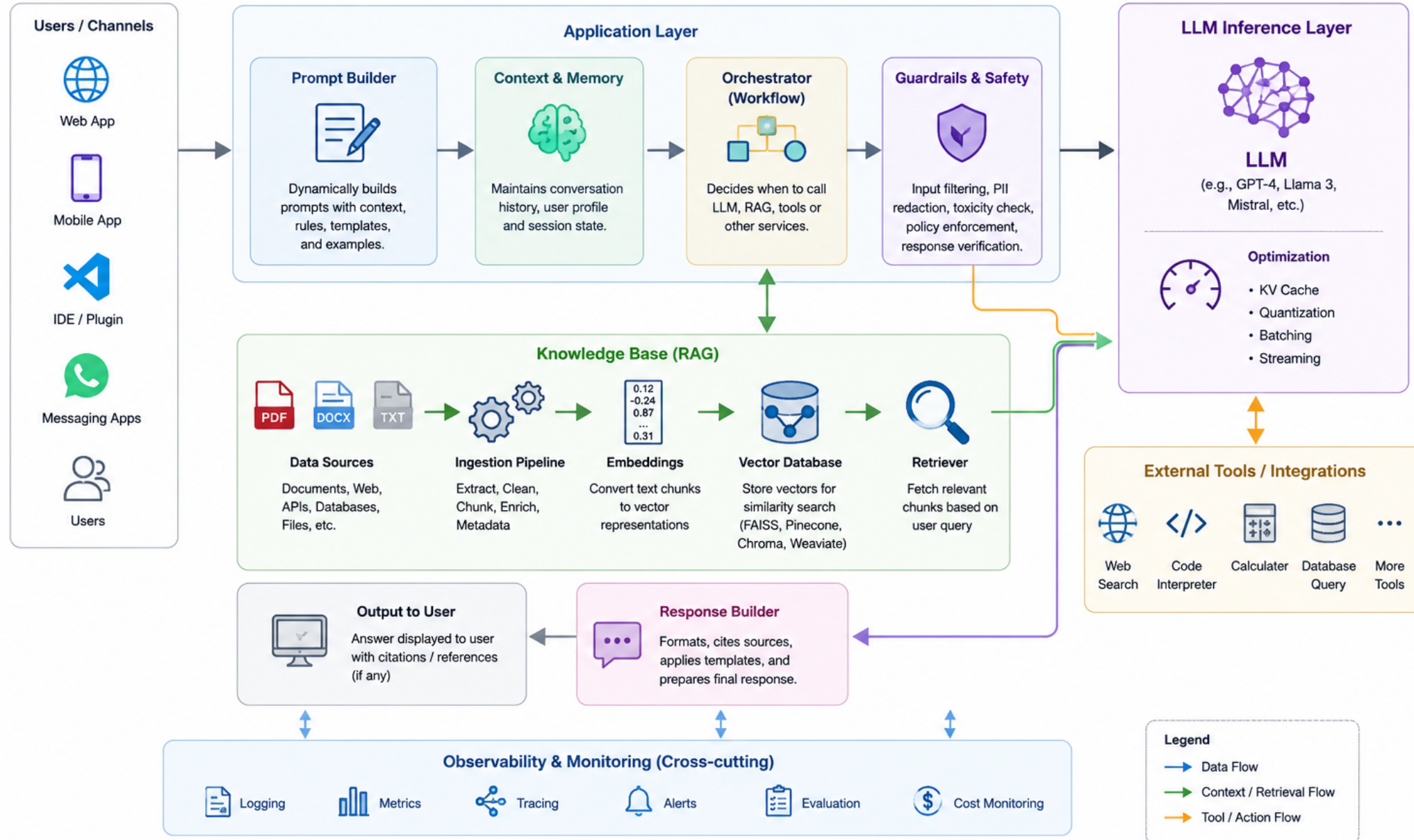
Neural networks trained on web-scale text to predict the **next token** in a sequence. Scale enables unexpected emergent behavior.

Statistical System: Operates probabilistically rather than querying a static relational database.

Emergent Abilities: Translation, reasoning, and summarization emerge from token prediction.

Anatomy of an LLM Application

Architecture / Block Diagram



Hospital AI Case Study

Stress-Testing "Just Use ChatGPT"

If a hospital asks for an AI assistant, sending raw queries to a base LLM creates critical failure points: no access to EHR records, inability to place lab orders, and zero compliance verification.

Production Requirements:

- EHR Database integration for authentic patient history.
- Medical policy retriever for accurate clinical guidance.
- Action APIs to schedule appointments & order lab tests.



Core Categories of LLM Apps



Conversational AI

Customer support bots, general chat assistants, and voice agents with persistent user memory and tone alignment.



Coding Tools

Inline code autocompletion, repository Q&A, and autonomous agentic coding frameworks like Claude Code.



Search & Research

Answer engines and Document Q&A (RAG) that cite factual source passages from enterprise repositories.

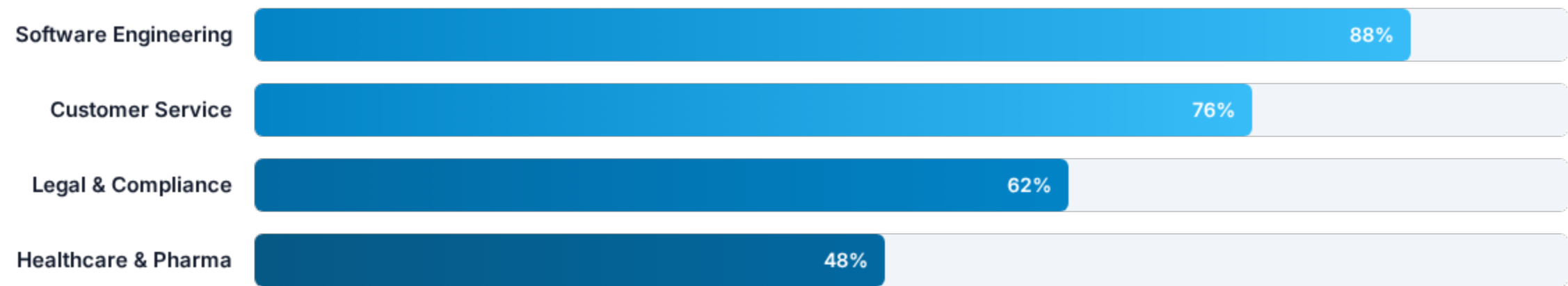
Key Application Modalities

-  **Interactive Assistants:** Combine UI chat, conversation memory buffers, and moderation layers to guide natural dialogue continuously.
-  **IDE & Developer Tools:** Leverage broad codebase context, file system tools, and terminal runners to automate software lifecycles.
-  **Document Intelligence:** Ingest massive PDFs, contracts, and filings to extract structured JSON data with zero human intervention.
-  **Autonomous Agents:** Equip models with planning frameworks and external APIs so they can execute multi-step workflows independently.

Why Apps Need Beyond an LLM

Base LLM Limitation	Required Application Layer	Practical Functionality
Hallucinations & Untruths	Retrieval Augmented Generation (RAG)	Grounds outputs in real-time retrieved facts.
No Knowledge of Private Data	Vector DBs & Embeddings	Allows search across internal documents.
Cannot Take Action	Tool / API Integration	Executes database updates and sends notifications.
Single Turn Memory	Session & Context State	Maintains chat history across turns.
Unsafe / Offensive Text	Input/Output Guardrails	Filters toxic prompts & validates security compliance.

LLM Adoption Across Sectors



Percent of enterprise organizations actively deploying LLM-backed applications in production environments.

Three Convergence Trends

3x

Synergistic Drivers

Why GenAI Exploded Now

- 1. Vast Digitized Data:** Trillions of tokens of high-quality human text and code became accessible for model training.
- 2. Massive GPU Compute:** Parallel computing acceleration enabled training models with hundreds of billions of parameters.
- 3. Transformer Architecture:** The 2017 self-attention mechanism unlocked scalable training over long context windows.

Key Takeaways & Next Steps

Core Takeaways

- **LLMs predict tokens:** Capabilities emerge from next-token optimization at scale.
- **LLM $\neq$ Application:** The model is the brain; product software supplies tools, data, and interfaces.
- **Top-Down Strategy:** Understanding what apps do precedes learning how transformers work.

Upcoming Lectures

- **Week 3: AI Agents:** Tool-use, planning loops, and multi-step autonomous behavior.
- **Week 4: RAG Fundamentals:** Vector databases, chunking strategies, and retrieval evaluation.
- **Weeks 5+:** Descending into tokenization, embeddings & transformer math.

Questions & Discussion

Lecture 1 & 2 Complete: Introduction to AI & LLM Applications